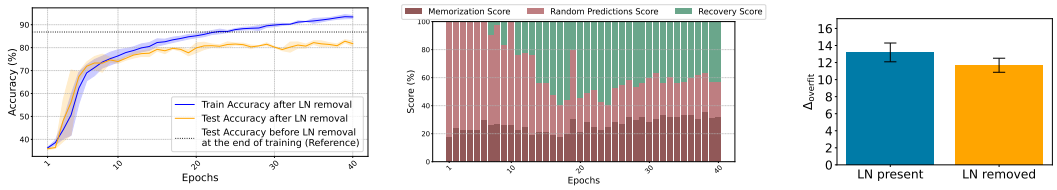
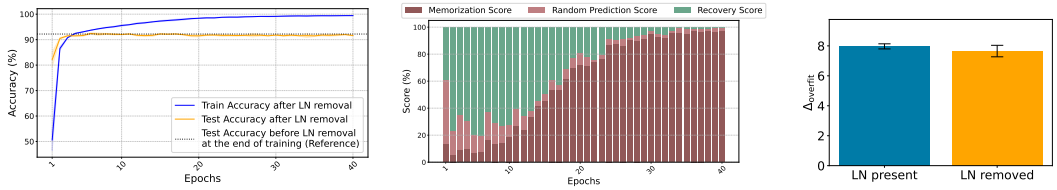


(a) Learning (Test) accuracy over epochs for Post-LN Model (RoBERTa)

(b) Memorization, Recovery and Random Predictions over epochs for Post-LN Model (RoBERTa)

(c) Overfitting gap for Post-LN Model (RoBERTa)

Figure 15: **LN removal suppresses memorization & facilitates true label recovery in Post-LN model - RoBERTa, TweetTopic Dataset:** LN removal in Longformer suppresses memorization and facilitates true label recovery, while minimal drop in learning. This further reduces overfitting seen by decreasing train-test accuracy gap when LN is removed. *Note: For RoBERTa we see a slight drop in learning accuracy which is much lower in comparison to the huge drop which happens in GPT2 setup in Fig. 7. Furthermore, memorization is still suppressed in comparison to GPT2 where memorization still persisted.*



(a) Learning (Test) accuracy over epochs for Post-LN Model (DistilBERT)

(b) Memorization, Recovery and Random Predictions over epochs for Post-LN Model (DistilBERT)

(c) Overfitting gap for Post-LN Model (DistilBERT)

Figure 16: **LN removal does not suppress memorization in DistilBERT (Emotions Dataset) but does not affect learning:** LN removal in DistilBERT does not suppress memorization (but it does not impact learning) as we observe for all other 5 Post-LN models. We think other components in the Transformer architecture might have a more profound impact on memorization in DistilBERT. But, since this work is primarily about LN impact, we refrain from dealing with other components, making it an interesting future work.

G.2 Significance of Early Layers LN

In this section, we illustrate the results corresponding to the significance of Early Layers LN in impacting learning and suppressing memorization for remaining Pre- and Post-LN models, respectively, across multiple datasets.

The results verified against Pre-LN models - GPTNeo, Qwen2, GPT2, RoBERTa-PreLayernorm, ViT-B, ViT-S, and Post-LN models - BERT, ELECTRA, Longformer, RoBERTa, DistilBERT are shown in Figs. 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, and 27.

G.2.1 Pre-LN Models: Early Layers LN drives learning

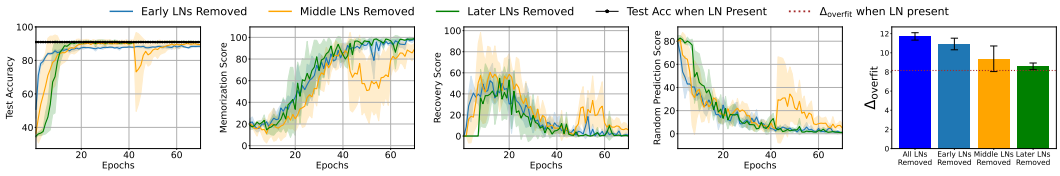


Figure 17: **Pivotal impact of early LNs for learning in Pre-LN model (GPTNeo, Emotions Dataset).** We can clearly observe the impact of early layers LN removal on destabilizing learning in GPTNeo, accompanied with higher train-test-accuracy gap, $\Delta_{\text{overfit}}^{\text{Pre, early}}$, than for later layers, $\Delta_{\text{overfit}}^{\text{Pre, later}}$, and poor memorization suppression.

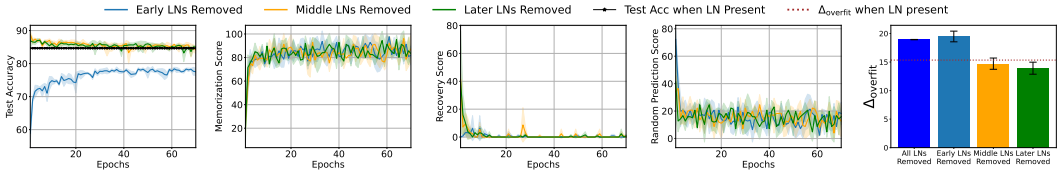


Figure 18: **Pivotal impact of early LNs for learning in Pre-LN model (Qwen2, News Dataset).** We can clearly observe the impact of early layers LN removal on destabilizing learning in Qwen2, accompanied with higher train-test-accuracy gap, $\Delta_{\text{overfit}}^{\text{Pre, early}}$, than for later layers, $\Delta_{\text{overfit}}^{\text{Pre, later}}$, and poor memorization suppression.

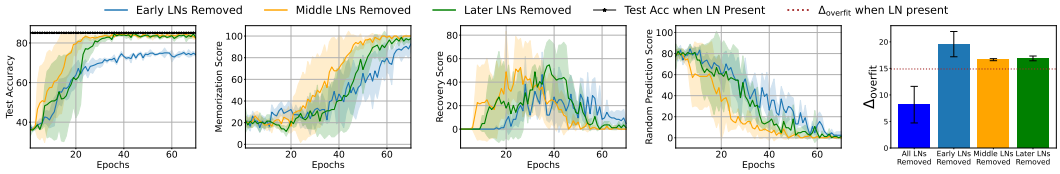


Figure 19: **Pivotal impact of early LNs for learning in Pre-LN model (GPT2, TweetTopic Dataset).** We can clearly observe the impact of early layers LN removal on destabilizing learning in GPT2, accompanied with higher train-test-accuracy gap, $\Delta_{\text{overfit}}^{\text{Pre, early}}$, than for later layers, $\Delta_{\text{overfit}}^{\text{Pre, later}}$, and poor memorization suppression. *Note: When all layers LNs are removed,* $\Delta_{overfit}^{Pre,\ all}$ *is low because when we removed all LNs from GPT2 then the model could not stabilize in training, reaching very low train accuracy, similar train-test accuracy as seen in Fig. 7a, hence low* $\Delta_{overfit}^{Pre,\ all}$*. However, when we just removed early LNs, the model is able to converge to a sufficiently high train accuracy, but the learning is impacted much severely in comparison to later layers.*

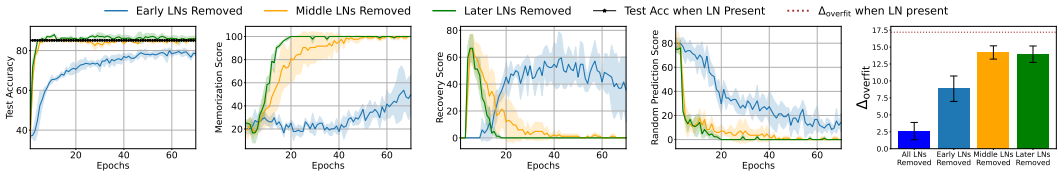


Figure 20: **Pivotal impact of early LNs for learning in Pre-LN model (RoBERTa-PreLayerNorm, TweetTopic Dataset).** We observe that for RoBERTa-PreLayerNorm, removing early layers LN impacts learning, however, memorization also seems to get suppressed, when compared with removing all LNs where we observe persistent memorization and destabilized learning (Fig. 11a). Despite this unique trend of RoBERTa-PreLayernorm (not following the consistent trend just as in the other 6 Pre-LN models), early layers LN still remain the most significant.

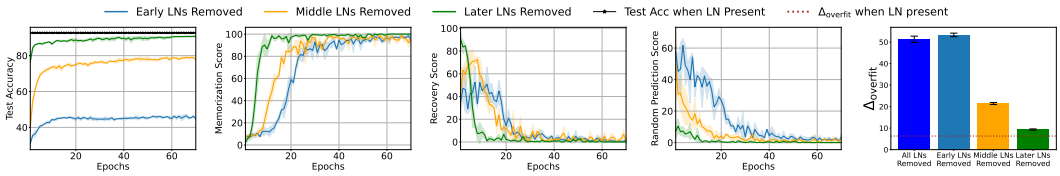


Figure 21: **Pivotal impact of early LNs for learning in Pre-LN model (ViT-B, CIFAR10 Dataset).** We can clearly observe the impact of early layers LN removal on destabilizing learning in ViT-B, accompanied with higher train-test-accuracy gap, $\Delta_{\text{overfit}}^{\text{Pre, early}}$, than for later layers, $\Delta_{\text{overfit}}^{\text{Pre, later}}$, and poor memorization suppression.

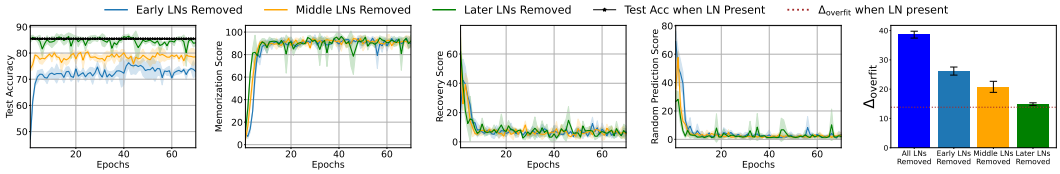


Figure 22: **Pivotal impact of early LNs for learning in Pre-LN model (CIFAR100, ViT-S Dataset).** We can clearly observe the impact of early layers LN removal on destabilizing learning in ViT-S, accompanied with higher train-test-accuracy gap, $\Delta_{\text{overfit}}^{\text{Pre, early}}$, than for later layers, $\Delta_{\text{overfit}}^{\text{Pre, later}}$, and poor memorization suppression.

G.2.2 Post-LN Models: Early Layers LN suppresses memorization

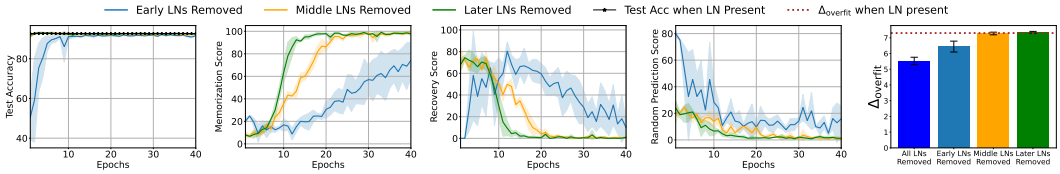


Figure 23: **Pivotal impact of early LNs on memorization in Post-LN model (BERT, Emotions Dataset).** We can clearly observe the impact of early layers LN removal on suppressing memorization & achieving true label recovery in BERT, accompanied with higher train-test-accuracy gap, $\Delta_{\text{overfit}}^{\text{Pre, early}}$, than for later layers, $\Delta_{\text{overfit}}^{\text{Pre, later}}$, while learning being intact.